

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
6 FT	(a)			Indicative content: Energy will transfer from the hot liquid to the cold surroundings. The plastic stopper/glass walls are both good insulators. These will significantly reduce the heat loss via conduction. The silver coatings are poor emitters and good reflectors of heat radiation / infra-red and will significantly reduce heat loss via radiation. The vacuum will prevent both conduction and convection losses as there are no particles. The stopper also stops any hot gases rising from the inside of the flask. This reduces heat transfer via convection. 5-6 marks Detailed description of how heat transfer is reduced by conduction, convection and radiation. <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured.</i> 3-4 marks Detailed description of how heat transfer is reduced by 2 out of conduction, convection and radiation. Alternatively, a brief correct description of all 3 methods. <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure.</i> 1-2 marks Detailed description of how heat transfer is reduced by 1 out of conduction, convection and radiation. Alternatively, a brief correct description of up to 2 of the methods. <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure.</i> Note. 3 marks AO1 description of process. 3 marks AO2 application to novel situation. 0 marks <i>No attempt made or no response worthy of credit.</i>	3	3		6		

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)			Correct for first 2 hours [as $\frac{10}{2} = 5^{\circ}\text{C}/\text{hour}$] (1) Incorrect for the whole 10 hours as $\frac{30}{10} = 3^{\circ}\text{C}/\text{hour}$ / incorrect because for the last 8 hours $\frac{20}{8} = 2.5^{\circ}\text{C}/\text{hour}$ (1) Alternative: After 2 hours temperature reduced by 10°C (1) after 10 hours should have reduced by 50°C / after 10 hours should have finished at 40°C / in the last 8 hours it should have dropped by 40°C so not always correct (1)			2	2	2	
				Question 6 total	3	3	2	8	2	0